

- d. Resume processing of any queued message that triggered synchronization according to [Section 4.6.7, “Counter Processing of Incoming Messages”](#).
 - i. If more than one message is queued from the synchronized peer, using the same operational group key, the messages SHALL be processed in the order received.
- e. Close the synchronization exchange created for the original *MsgCounterSyncReq* message.

4.18.6. Message Counter Synchronization Session Context

While conducting Message Counter Synchronization with a peer, nodes SHALL maintain the following session context. For nodes initiating message counter synchronization, this context SHALL be maintained throughout the full exchange of *MsgCounterSyncReq* and *MsgCounterSyncRsp* messages. For nodes responding to *MsgCounterSyncReq* messages, the context SHALL only be maintained long enough to generate and successfully transmit the *MsgCounterSyncRsp* message.

1. **Fabric Index**: Records the [Index](#) for the Fabric within which nodes are conducting message counter synchronization.
 - Fabric Index is derived by identification of an Operational Group Key associated with the fabric through successful decryption with that key and verification of the [Message Integrity Check](#). For nodes initiating counter synchronization, this occurs at decryption of an inbound groupcast message. For nodes in the responder role, this occurs at decryption of an inbound *MsgCounterSyncReq* message.
2. **Peer Node ID**: Records the node ID of the peer with which message counter synchronization is being conducted.
 - For nodes initiating message counter synchronization, this is the node ID of the responder. For nodes responding to message counter synchronization, this is the node ID of the initiator.
3. **Role**: Records whether the node is the initiator of or responder to message counter synchronization.
 - Together, *Fabric Index*, *Peer Node ID* and *Role* comprise a unique key that can be used to match incoming messages to ongoing MCSP exchanges.
4. **Message Reception State**: Provides tracking for the [Control Message Counter](#) of the remote peer.
5. **Peer IP Address**: The unicast IP address of the peer.
6. **Peer Port**: The receiving port of the peer.
7. **Operational Group Key**: The [Operational Group Key](#) that is being used to encrypt messages within the counter synchronization exchange.
8. **Group Session ID**: Records the [Group Session ID](#) derived from the [Operational Group Key](#) that is being used to encrypt messages within the counter synchronization exchange.

4.18.7. Sequence Diagram

The following sequence diagram shows an example of how message counter synchronization behaves in the most common scenario.